Summary of Bellman's *The Theory of Dynamic Programming* (1954)

Overview

Bellman's classic paper introduces the *principle of optimality*, proposing that many complex decision problems whether deterministic or stochastic, discrete or continuous—can be decomposed into simpler subproblems. This decomposition allows solving large-scale problems recursively.

1. Discrete Deterministic Case

Mathematical Formulation:

$$f_N(p) = \max_k f_{N-1}\left(T_k(p)\right)$$

The state transitions deterministically based on the selected transformation T_k. The aim is to find a sequence of actions that maximizes the final return.

Example: Choose a route deterministically to maximize total reward.

2. Discrete Stochastic Case

Mathematical Formulation:

$$f_N(p) = \max_k \int f_{N-1}(z)\, dG_k(p, z)$$

Here, the transformation is stochastic. The function $G_k(p, z)$ gives the probability distribution over next states z after choosing action k in state p.

Example: A robot moving on uncertain terrain with probabilistic outcomes for each action.

3. Continuous Deterministic Process

Mathematical Formulation:

$$f(p; S + T) = \sup_{u(\cdot) \in \mathcal{D}[0,S]} f\left(x^u(S); T\right)$$

This setting models continuous-time decision-making. As $S \to 0$, the functional equation leads to a partial differential equation, such as the Hamilton-Jacobi-Bellman equation.

Example: Controlling vehicle speed continuously to maximize distance traveled within a fixed time.

Examples in the Paper

The paper references problems from the calculus of variations, resource allocation over time, and multi-stage decision-making under uncertainty, such as inventory control and search problems.

Conclusion

Bellman's formulation unifies a wide class of decision-making problems under a single recursive structure. Whether the system is deterministic or stochastic, discrete or continuous, the key insight is that optimal strategies can be derived from solving simpler subproblems step by step.